

INTERNSHIP REPORT

On

BEAST OS

A Cross-Platform EdTech Operating System by Beast Learners

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.



By

Roll Number of Student: **CS-2341737**

Name of Student: **SUSHANT SINGH**

Section: **4CSE10**

Batch: **2023–27**

CANDIDATE'S DECLARATION

I, Sushant Singh, hereby declare that this internship report titled "Beast OS – A Cross-Platform EdTech Operating System for ICSE Board Preparation" is based on my own work carried out during my internship at Beast Learners. I have described, in my own words, the modules I was responsible for and the challenges I encountered while building them. This report has not been submitted elsewhere for any other degree or diploma requirement, and I take full responsibility for its originality.

Signature: _____

Student Name: Sushant Singh

Roll No.: CS-2341737

Date: 15/09/2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the engineering team at Beast Learners for the opportunity to contribute to a live, student-facing product during my internship. Their guidance on component structuring, state management, and dashboard design helped me understand how front-end decisions translate into real usability for students preparing for board exams.

I am also thankful to the Department of Computer Science and Engineering, IILM University, Greater Noida, for making this internship possible as part of the academic programme, and to my mentors on the team who patiently reviewed my early implementations of the dashboard and admin screens.

Finally, I thank my family and peers, including fellow interns on the BeastOS team, for their support through the internship.

Signature: _____

Student Name: Sushant Singh

Roll No.: CS-23411294

INTERNSHIP COMPLETION CERTIFICATE

BEAST LEARNERS

Zeta-1, Greater Noida, 201306

+91 9102132866
careers@beastlearners.com



INTERNSHIP COMPLETION CERTIFICATE

September 5, 2026

This is to certify that **Sushant Singh** has successfully completed his three-month internship as a **Web Developer Intern** at **Beast Learners**, from **June 1, 2026 to September 1, 2026**.

During his internship, he contributed to the **development and enhancement of BeastOS**, the organization's digital platform, working on web development, feature implementation, and technical improvements. He demonstrated dedication, technical skills, and a strong willingness to learn throughout his internship.

We appreciate his valuable contributions to **Beast Learners** and wish him continued success in his future endeavors.

Sincerely,

Aditya Raj
Founder
Beast Learners

TABLE OF CONTENTS

- Candidate's Declaration — 2
- Acknowledgement — 3
- Internship Completion Certificate — 4
- Table of Contents — 5
- 1. Project Description — 6
 - 1.1 Introduction
 - 1.2 Organization Profile
 - 1.3 Problem Statement
 - 1.4 Project Objectives
 - 1.5 Scope of the Project
 - 1.6 Technologies and Tools Used
 - 1.7 System Architecture
 - 1.8 Methodology
 - 1.9 Expected Outcomes
 - 1.10 Certificates of Completion and Communication Proof
- 2. Bibliography / References

1. PROJECT DESCRIPTION

1.1 Introduction

This report documents the internship completed by Sushant Singh, a B.Tech Computer Science and Engineering student (Section 4CSE10) at IILM University, Greater Noida, U.P., at Beast Learners — an EdTech company building tools for ICSE-level board exam preparation. The internship was undertaken in the role of Web Developer Intern, with hands-on responsibility for two areas of the flagship product, Beast OS: the student-facing entry experience (Home Dashboard and AI Tutor interface) and the internal Admin/B1 reporting layer, including the Subscription Store.

Beast OS reframes exam preparation as a single connected platform rather than a set of disconnected study apps — a student logs in once and gets a live view of their streak, an AI tutor, quiz tools, and a readiness tracker, while the organisation gets an equally live view of usage, subscriptions, and credit consumption on the admin side. My work during the internship sat at both ends of this: the first screen a student sees, and the dashboard the organisation uses to understand what students are actually doing.

1.2 Organization Profile

Beast Learners is an EdTech company built around Beast OS, a platform positioned as an "operating system" for board-exam preparation rather than a static content library. The product combines syllabus tracking, AI-assisted tutoring, quiz generation, and a credit-based monetisation model into one cohesive web application.

The engineering culture at Beast Learners is fast-moving and component-driven: the front end is a single React.js + Tailwind CSS application, with a mirrored but access-restricted Admin Dashboard for internal teams. Interns are onboarded directly onto production screens rather than sandbox exercises, which meant my work — the login-first Home screen and the internal analytics panels — was visible to real users and the internal team from day one.

1.3 Problem Statement

Before Beast OS, ICSE board preparation for most students was split across printed guides, generic quiz apps, and tutoring that had no memory of what a student had already studied. From the specific angle of the modules I worked on, the gaps were:

- No welcoming, motivating "home base" for students — most study apps open straight into a feature, with no sense of streaks, daily goals, or momentum.
- AI tutoring tools that either had no visible cost/credit system (leading to overuse) or no clear onboarding message for first-time use.
- No clean, tiered way for students to upgrade to a paid plan, with pricing and features often buried in settings menus.
- No internal dashboard for the organisation to see, at a glance, how many students were actively subscribed, how credits were being consumed, and which plan was most popular — meaning product decisions were being made without real usage data.

Beast OS addresses these by giving students a warm, streak-driven Home screen and a clearly priced Subscription Store, while giving the internal team a live BI dashboard to monitor exactly this activity.

1.4 Project Objectives

- To design and build the Student Home Dashboard — the first screen shown after login — combining a personalised greeting, a daily challenge banner, a study-streak counter, and quick-access navigation cards.
- To build the front-end shell of the AI Tutor interface, including its credit-balance display and welcome-state messaging.
- To design and build the Subscription Store, presenting Monthly and Yearly plans with clear feature lists and an incentive to choose the annual plan.
- To contribute to the internal Admin BI Analytics Dashboard, translating raw usage numbers (active subscribers, credits burned, daily active users) into readable stat cards and charts.
- To gain practical experience in component reuse across very different audiences — the same visual language had to work for a student on mobile and an admin viewing analytics on desktop.

1.5 Scope of the Project

My scope covered the front-end implementation of the Home Dashboard, the AI Tutor interface shell, the Subscription Store, and sections of the Admin BI Dashboard — specifically the top-line stat cards and the two charts (Daily Active Credits, Subscription Plan Distribution). This involved React component design, Tailwind responsive layout, and wiring components to already-existing backend endpoints so the numbers shown were live rather than hardcoded.

Work on the Quiz Generator and the Command Centre readiness screens was carried out by other members of the intern team; I refer to them here only for platform context, not as part of my personal contribution. Backend logic, the credit-deduction system, payment processing, and database design were likewise outside my scope and owned by the full-stack team.

1.6 Technologies and Tools Used

Table 1.1: Technology Stack Used in Beast OS (Web Developer Scope)

Layer	Technology / Tool	Purpose
Frontend (Web)	React.js, Tailwind CSS	Building the Home Dashboard, AI Tutor shell, Subscription Store, and admin stat cards
Data Visualisation	Recharts	Rendering the Daily Active Credits line chart and Subscription Plan Distribution bar chart on the Admin panel
State & Data Fetching	React Hooks, REST API integration	Pulling live streak, credit, and subscriber data into components
UI/Icons	Lucide icons, custom card components	Consistent look across Home, AI Tutor, and Subscription screens

Backend (consumed, not built)	Supabase (PostgreSQL, Edge Functions)	Source of streak data, credit balances, and subscriber counts
Payments (consumed, not built)	Razorpay API	Powering the "Upgrade Now" actions on the Subscription Store

1.7 System Architecture

Beast OS uses a decoupled architecture: the React.js client is a pure presentation layer that reads from and occasionally triggers actions on a shared backend via REST APIs. My part of this system sat at the two "edges" of the user journey — the very first screen a logged-in student sees (Home), and the internal reporting screen that summarises what all students are doing in aggregate (Admin BI Dashboard) — with the Subscription Store connecting the two, since upgrade decisions made by students there are what the Admin dashboard's subscriber charts ultimately reflect.

All components I built were designed to be stateless where possible, receiving readiness numbers, streak counts, and subscriber totals as props from a data-fetching layer rather than calculating anything client-side, so that the Home screen a student sees and the numbers an admin sees always trace back to the same backend source of truth.

Figure 1.1: Front-End Data Flow for Home Dashboard, Subscription Store, and Admin BI Dashboard

1.8 Methodology

Work proceeded module by module, with each screen wireframed, built with reusable components, connected to live data, and reviewed before merge.

1.8.1 Student Home Dashboard

- Built the "Hello, Beast!" welcome header and the Daily Challenge banner, which surfaces the student's current weakest topic (e.g., "Physics – Force") as a call-to-action.
- Implemented the Study Streak widget, including its zero-state ("0 Days") so new users don't see a broken-looking counter on their first login.
- Built the four-card quick-navigation grid (AI Tutor, Quiz, Notes, Library) as a single reusable card component, and the Syllabus Progress section with its own empty-state fallback for students who haven't started a course yet.

1.8.2 AI Tutor Interface

- Built the entry screen for the AI Tutor — the "Beast AI Tutor" header, the welcome message bubble, and the message input bar.
- Implemented the visible credit counter (e.g., "500 Credits") in the header, so the cost of AI usage is transparent to the student before they start chatting.
- Handled the empty conversation state so the screen doesn't look unfinished before a student sends their first message.

1.8.3 Subscription Store

- Built the two-plan comparison layout — Pro Monthly (₹99/month, recurring, cancel anytime) and Pro Yearly (₹599 total, one-time payment, no autopay) — with matching feature checklists (AI credits, unlimited tutor access, downloadable notes).

- Added the savings badge on the annual plan ("Save ₹589 (₹50/mo)") to visually nudge students toward the better-value option without hiding the monthly plan.
- Ensured both plan cards used the same underlying component so future plan changes wouldn't require rebuilding the layout.

1.8.4 Admin BI Analytics Dashboard

- Built the three top-line stat cards — Active Subscribers, Credits Burned (Today), and Live DAU — as a single reusable stat-card component driven by an icon, label, and value.
- Implemented the Daily Active Credits line chart (last 7 days) and the Subscription Plan Distribution bar chart (Elite Plus / Elite Basic / Pro Monthly) using Recharts, wired to live REST data rather than static numbers.
- Coordinated with the team members responsible for the Command Centre and Quiz Generator to keep the shared card/chart component library consistent across both the student-facing and admin-facing sides of the product.

1.8.5 Cross-Cutting Challenges

Table 1.2: Key Technical Challenges and Solutions

Challenge	Solution Implemented
The same stat-card pattern needed to work for very different audiences — a student's streak counter and an admin's "Active Subscribers" figure — without looking like two different design systems.	Built one parameterised stat-card component (icon, label, value, colour) reused across the Home Dashboard and the Admin BI Dashboard.
New users would see the Home Dashboard with a "0 Days" streak and "No progress records found," which risked looking like a bug.	Designed explicit, on-brand empty states for every widget instead of leaving blank or default browser text.
The Subscription Store needed to make the yearly plan attractive without making the monthly plan look like a bad option.	Used a distinct highlighted badge and framing ("one-time payment, no autopay") on the yearly card rather than dimming or shrinking the monthly card.
Admin chart data (credits burned, DAU) could legitimately be zero on a quiet day, and a flat zero-line chart risked looking like the chart had failed to load.	Kept axis scaling and gridlines visible regardless of data range, so a flat line at zero still reads as "no activity" rather than "broken chart."
Credit balance in the AI Tutor header needed to stay accurate without a manual refresh, especially right after a student used the Daily Challenge or Quiz.	Connected the credit display to the same REST endpoint used by the backend's credit-deduction logic, so it re-fetches on page load rather than relying on a stale local value.

1.9 Expected Outcomes

By the end of the internship, four front-end modules I was responsible for were live in production: the Home Dashboard, the AI Tutor interface shell, the Subscription Store, and a section of the Admin BI Dashboard. Together these cover the full loop from a student's first login, through their decision to upgrade, to the organisation's ability to see that decision reflected in real data.

Personally, this internship gave me direct experience building components that had to serve two very different users — a student and an internal admin — from the same underlying design system, along

with practical exposure to Recharts-based data visualisation, empty-state design, and consuming live REST APIs rather than working against static mock data. This experience has strengthened my readiness for a professional front-end/UI engineering role.

As future scope, the Home Dashboard's Daily Challenge could be extended to show a short history of past challenges, the Subscription Store could support a mid-tier "Elite Basic" plan alongside Pro, and the Admin BI Dashboard could add date-range filtering to the existing charts.

1.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by Beast Learners is attached in the front matter of this report. Screenshots and other communication proof from the internship period are appended below.



2. BIBLIOGRAPHY / REFERENCES

- [1] React – A JavaScript library for building user interfaces. Available at: <https://react.dev>
- [2] Tailwind CSS Documentation. Available at: <https://tailwindcss.com/docs>
- [3] Recharts Documentation. Available at: <https://recharts.org>
- [4] Supabase Documentation – Database, Edge Functions and Realtime. Available at: <https://supabase.com/docs>
- [5] PostgreSQL 16 Documentation. Available at: <https://www.postgresql.org/docs/>
- [6] Razorpay API Reference – Subscriptions and Orders. Available at: <https://razorpay.com/docs/api/>
- [7] Lucide Icons Documentation. Available at: <https://lucide.dev>
- [8] MDN Web Docs – Responsive Web Design Basics. Available at: <https://developer.mozilla.org>